

# Informative Power-System Probing via Sequential Bayesian Experimental Design

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**Abstract**—Uncertain inertia and frequency-response coefficients complicate the characterization of power-system dynamics. We formulate sequential probing as Bayesian experimental design: pulse durations are selected from noisy measurements, and the physical state is carried between probes. This coupling makes each duration relevant to both the current observation and subsequent experiments. We compare six methods on reduced IEEE 9-bus and IEEE 14-bus swing models under common sensing, duration constraints, and probe budgets. Deep adaptive design (DAD), Step-DAD, and reinforcement-learning-based sequential design (RL-sBOED) achieve higher mean sequential prior-contrastive (sPCE) information scores than fixed-sequence, myopic, and random designs. On IEEE 9-bus, DAD improves on the optimized Fixed design by 0.09–0.13 nats. DAD and RL-sBOED require only policy evaluation online, whereas Step-DAD adds online refinement and achieves modest gains on IEEE 9-bus. Rescoring IEEE 14-bus histories with up to  $10^5$  contrasts preserves the learned-policy advantage but reverses the Fixed-Myopic ordering at longer horizons. Scores remain sensitive to the contrast count, limiting fine rankings among learned policies. These results characterize the information–computation tradeoff of sequential probing within the specified benchmark. Code and saved results are available at [https://github.com/alexschmidt123/tpc2027\\_power\\_grid](https://github.com/alexschmidt123/tpc2027_power_grid).

**Index Terms**—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

## I. INTRODUCTION

As inverter-based resources displace synchronous generation, changing inertia and frequency response increase the need to characterize grid dynamics [1]. Measurement-based inertia estimation uses frequency and active-power responses to disturbances [2]; controlled probing offers an alternative when naturally occurring events do not provide suitable excitation. Field tests on the western North American interconnection demonstrate that probing-signal design affects the quality of electromechanical-mode estimates [3], and recent work applies probing to inertia and frequency-response estimation in systems with inverter-based resources [4].

Informative excitation is a central problem in system identification [5]. We consider how successive pulse durations should adapt to measurements when the system is not reset between probes. A longer pulse changes the injected energy,

the measured transient, and the initial state of the next experiment. Consequently, optimizing each probe in isolation can overlook its effect on later measurements. The design must account for the entire probing sequence while using only the observations available at each decision.

Sequential Bayesian optimal experimental design provides a framework for this tradeoff [6]. A fixed sequence cannot respond to incoming measurements, whereas a myopic design optimizes only the next observation. Deep adaptive design (DAD) [7] and reinforcement-learning-based sequential design (RL-sBOED) [8] learn history-dependent policies using a full-sequence information objective. Step-DAD [9] additionally refines a pretrained policy during the experiment, trading online computation for further adaptation.

This paper develops a Bayesian duration-selection formulation for reduced IEEE 9-bus and IEEE 14-bus models with uncertain machine inertia and effective frequency response. It adapts six existing design methods to a common simulator that carries the physical state between probes, and compares their information scores, selected durations, and computational costs. The comparison distinguishes observation-dependent policies from fixed designs and full-sequence planning from one-step planning. All results concern the same constrained duration space and probe budget.

## II. PROBLEM FORMULATION

The objective is to reduce uncertainty in grid parameters using  $T$  sequential probes. The index  $t = 1, \dots, T$  denotes the design stage,  $d$  denotes physical time within a probe window, and  $D_t$  is the selected pulse duration.

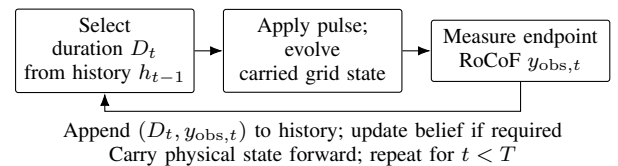


Fig. 1. Sequential probing loop. Adaptive methods use observation history; Fixed and Random select durations without observation feedback.

TABLE I  
NOMINAL PARAMETER SCALES AND FIXED DAMPING

| Network     | Bus     | $\bar{M}_i$ | $\bar{K}_i$ | $D_i$     |
|-------------|---------|-------------|-------------|-----------|
| IEEE 9-bus  | 1       | 0.1254141   | 0.2937500   | 0.0125414 |
|             | 2       | 0.0339531   | 0.1312500   | 0.0033953 |
|             | 3       | 0.0159685   | 0.0750000   | 0.0015969 |
| IEEE 14-bus | 1       | 0.0212207   | 0.0784314   | 0.0021221 |
|             | 2       | 0.0344836   | 0.1274510   | 0.0034484 |
|             | 3, 6, 8 | 0.0265257   | 0.0980392   | 0.0026526 |

### A. Latent Parameters and Power-System Model

For a reduced network with  $m$  machines, the latent parameter vector is  $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$ , where  $M_i$  and  $K_i$  are inertia and effective frequency-response coefficients. The prior  $p_0(\theta)$  has independent components  $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$  and  $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$ . One draw of  $\theta$  remains fixed throughout all  $T$  probes.

Let  $\delta_i$  be rotor angle,  $\omega_i$  angular-speed deviation, and  $u_i$  the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left( D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to  $d$ . In (1),  $M_i$  scales acceleration and  $K_i$  models the power response to frequency deviation; no explicit turbine–governor dynamics are included. The damping  $D_i$  and couplings  $B_{ij}$  are known, and  $\Delta f_i = \omega_i/(2\pi)$  is measured in Hz. On a 100-MVA base at 60 Hz, the units of  $M_i$ ,  $K_i$ , and  $D_i$  are pu s<sup>2</sup>/rad, pu/Hz, and pu s/rad. Angles, speed deviations, and net mechanical injections initially equal zero, giving an unforced equilibrium.

The reduced IEEE 9-bus model retains buses 1, 2, and 3; the reduced IEEE 14-bus model retains buses 1, 2, 3, 6, and 8. Test-case reactances [10] define a lossless Laplacian  $\mathbf{L}$ . Eliminating algebraic buses  $l$  while retaining machines  $g$  gives  $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$ , with  $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$  for  $i \neq j$  and  $B_{ii} = 0$ . Resistance, voltage dynamics, and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as  $\bar{M}_i = 2H_i/(2\pi f_0)$ , with  $f_0 = 60$  Hz. For IEEE 9-bus,  $H = (23.64, 6.40, 3.01)$  s matches published machine data [11, Table 2]. For IEEE 14-bus,  $H = (4, 6.5, 5, 5, 5)$  s is prescribed at buses (1, 2, 3, 6, 8) as assumed benchmark values. For both networks,  $\bar{K}_i = K_{\text{tot}} w_i$ , with  $K_{\text{tot}} = 0.5$  pu/Hz. The allocation weights are  $w = (0.5875, 0.2625, 0.15)$  for IEEE 9-bus and  $w = (8, 13, 10, 10, 10)/51$  for IEEE 14-bus. Fixed damping satisfies  $D_i = (0.1 \text{ s}^{-1})\bar{M}_i$ . The response scales, damping values, and prior ranges are benchmark assumptions rather than validated operational grid parameters.

### B. Probe Design and Observation Model

The probe design is  $\xi_t = (A_t, B_t, D_t)$ : amplitude, bus location, and probe signal duration. This study fixes  $A_t = 0.05$  pu and  $B_t = 1$ ; only  $D_t$  is optimized. The raised-cosine input for  $0 \leq d \leq D_t$  is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero after  $D_t$  and at all other buses. Its injected energy is proportional to  $A_t D_t/2$ ; equal probe counts therefore do not imply equal energy budgets. The observation window is fixed at  $D_{\text{obs}} = 3.5$  s. The angle and speed state at its end initializes the next window without resetting the system. Thus, each response depends on the complete design sequence  $\xi_{1:t} = (\xi_1, \dots, \xi_t)$ , even though only duration is optimized.

Write  $\Delta f_{1,t}(d; \theta, \xi_{1:t})$  for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate of change of frequency (RoCoF), denoted by  $y_t$ :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments  $\theta, \xi_{1:t}$  are suppressed in the numerator. With sampling interval  $\Delta d_{\text{samp}} = 0.025$  s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where  $\sigma = 0.005$  Hz/s and measurement noise is independent across windows. This sensor measures the post-pulse transient rather than the initial inertial response. Its fixed location and scalar output provide a common observation model for all methods. Observations become available at the end of each window; decision latency is idealized as zero in the simulated dynamics.

### C. Sequential Decisions and Bayesian Updating

The data after step  $t$  are the ordered design–observation history  $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$ , with  $h_0 = \emptyset$ . A policy  $\pi$  selects  $\xi_t$  using only observed history  $h_{t-1}$ , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\text{min}}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

Here  $\Delta D_{\text{min}} = 0.01$  s. The lower bound is 0.2 s at the first stage and  $D_{t-1} + \Delta D_{\text{min}}$  thereafter; the upper bound is  $3 - (T - t)\Delta D_{\text{min}}$ , reserving room for the remaining probes. All methods map a unit action continuously into this interval, without sorting durations. Let  $\Pi$  be the resulting policy class. Increasing durations are a benchmark restriction, not a requirement of the swing equations; nonmonotonic sequences are outside this comparison. Policy randomness is independent of  $\theta$ , and actions depend only on observed history.

The Gaussian likelihood  $p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t)$  has mean  $y_t(\theta, \xi_{1:t})$  and variance  $\sigma^2$ . Bayes' rule updates parameter uncertainty after each measurement:

$$p(\theta \mid h_t) = \frac{p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t) p(\theta \mid h_{t-1})}{p(y_{\text{obs},t} \mid h_{t-1}, \xi_t)}. \quad (6)$$

The denominator integrates the numerator over  $\theta$ , with  $p(\theta | h_0) = p_0(\theta)$ . Each candidate parameter vector propagates its own physical state under the observed designs. Since the policy uses only observed history and independent auxiliary randomness, its action probabilities cancel from the Bayesian update.

#### D. Expected Information Gain and Design Objective

The expected information gain (EIG) of policy  $\pi$  measures the mean Kullback–Leibler divergence between the terminal posterior and the prior:

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T|\pi} [D_{\text{KL}}(p(\theta | h_T) \| p_0(\theta))]. \quad (7)$$

Here  $D_{\text{KL}}(p\|q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$ ; information gain is measured in nats. The expectation averages over parameter draws, measurement noise, and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Equation (8) is the common design objective. The methods differ in their approximation to this objective and their use of observation feedback, while sharing the same probe budget. Their computational costs are assessed separately.

Direct EIG evaluation requires marginalizing the nonlinear likelihood over uncertain parameters. We use the sequential prior-contrastive (sPCE) lower bound [7] as a tractable training and evaluation objective. Let  $\theta^{(0)} \sim p_0$  generate the observed history, and draw independent contrasts  $\theta^{(1)}, \dots, \theta^{(L)} \sim p_0$ . For any candidate parameter vector, its likelihood along the realized sequence is

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} | \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. Averaging  $\hat{I}_T$  estimates the sPCE lower bound on  $\mathcal{I}_T(\pi)$ , not the exact EIG. Individual scores may be negative but are bounded above by  $\log(L+1)$ . With  $L = 128$ , the ceiling is 4.8598 nats; near-ceiling scores can conceal differences in EIG. We report these estimated lower-bound values as information scores throughout the results. The policy does not receive the generating parameters.

### III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). Adaptation and planning horizon are distinct: an adaptive method can be myopic, whereas a fixed sequence can be optimized jointly for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

#### A. Nonadaptive Designs: Random and Fixed

Random samples  $z_t \sim \mathcal{U}(0,1)$  independently at each step and sets  $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$ . Here  $D_1^{\min} = 0.2$ ,  $D_t^{\min} = D_{t-1} + \Delta D_{\min}$  for  $t > 1$ , and  $D_t^{\max} = 3 - (T-t)\Delta D_{\min}$ . Thus, random samples are uniform within the remaining feasible interval and are not uniform over feasible complete sequences. Random does not use observations and needs no offline training.

Fixed is a nonadaptive Bayesian design: all durations are optimized jointly before observations are collected [6, Sec. 2.4]. The ordered sequence is optimized using the full-sequence sPCE objective. It compares short, central, long, and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

#### B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [6, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds, and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

#### C. Offline-Trained Adaptive Methods: DAD and RL-sBOED

DAD and RL-sBOED are trained on simulated sequences drawn from the parameter prior. Policy inputs contain the stage index, ordered past durations, scaled observations, and availability masks; future slots are masked. Online decisions require a policy forward pass and the feasible-duration map, without explicit posterior updates or planning simulations.

DAD [7] uses two 64-unit tanh hidden layers, a trainable stage bias, and a sigmoid action map. Training maximizes the terminal sPCE score using pathwise gradients through observations and carried states. Validation selects either random initialization or initialization from the separately trained Fixed sequence. The complete preparation cost includes this Fixed training stage (Section V-C).

RL-sBOED [8] uses a REDQ-style actor-critic algorithm with ten critics and five replay updates per trajectory batch. The actor has two 128-unit ReLU hidden layers. Rewards are  $R_t = \hat{I}_t - \hat{I}_{t-1}$ , with  $\hat{I}_0 = 0$  and discount factor  $\gamma = 1$ , so their sum equals the terminal sPCE score. Training uses entropy regularization in normalized tanh-action coordinates with target entropy  $-1$ . Evaluation uses the deterministic actor output; reported information scores exclude the entropy bonus.

#### D. Online Refinement: Step-DAD

Step-DAD [9] starts each episode from its selected DAD checkpoint and leaves the first probe unchanged. Before subsequent decisions, a 128-particle posterior supplies candidate parameters and their carried states for simulations of the remaining sequence. A fresh Adam optimizer performs four updates per decision (batch size 32, learning rate  $10^{-3}$ ). A fixed conditional validation batch selects the best candidate, including the unchanged policy. Retained weights persist within an episode only. The total refinement budget is  $4(T - 1)$  updates. Myopic and Step-DAD require posterior assimilation only before subsequent decisions; terminal information scoring is separate.

### IV. EXPERIMENTAL SETTINGS

#### A. Simulation and Training Protocol

We evaluate horizons  $T = 3, 4, 5$ , with one scalar observation per probe. Each network has separately trained policies; sensor settings and duration constraints are shared.

The swing equations are integrated by fourth-order Runge-Kutta (RK4) with step size  $1/640$  s. Pathwise gradients use numerical state and duration Jacobians: coordinate perturbations are  $10^{-5} \max(1, |x_j|)$  for state  $x_j$  and  $10^{-5}$  s for duration, clipped at the bounds. The validation suite checks policy and conditional-refinement gradients against end-to-end finite differences.

For each network and horizon, Fixed, DAD, and RL-sBOED are trained with seeds 101, 202, and 303. Training uses 2000 trajectory updates with batch size 32. Fixed and DAD use Adam with learning rate  $10^{-3}$ ; the RL actor and critics use  $3 \times 10^{-4}$ . Checkpoints are selected using 128 validation systems every 100 updates. Training, validation, and the initial information evaluations use  $L = 128$  contrasts; Section V-D reports additional IEEE 14-bus evaluations at larger  $L$ . Myopic and Step-DAD maintain 128 planning particles; Step-DAD performs four refinement updates before each decision after the first.

#### B. Evaluation Protocol

Testing uses seeds 1001, 1002, and 1003, with 512 episodes per seed, giving 1536 episodes per trained policy. Parameter and measurement-noise draws are paired across methods, while planners use a separate random stream. Training, checkpoint validation, and testing use separate seeds. For trained methods, the tables report mean  $\pm$  sample standard deviation (SD) across three training-seed means, each averaged over all test episodes. Random and Myopic are independent of training seed; their SD is calculated across the three test-seed means. These SDs describe seed variation, not episode dispersion or confidence intervals for method differences.

### V. RESULTS AND DISCUSSION

#### A. Information Acquired at Equal Probe Budgets

Tables II and III report sPCE information scores using the aggregation defined in Section IV-B. All three learned adaptive

methods exceed Random, Fixed, and Myopic in mean score for each network and horizon.

TABLE II  
IEEE 9-BUS sPCE SCORES ( $L = 128$ ; NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.4321 \pm 0.0310$ | $0.4924 \pm 0.0342$ | $0.5392 \pm 0.0471$ |
| Fixed    | $2.0112 \pm 0.0116$ | $2.3192 \pm 0.0004$ | $2.5066 \pm 0.0013$ |
| Myopic   | $1.9736 \pm 0.0438$ | $2.1707 \pm 0.0366$ | $2.2407 \pm 0.0547$ |
| DAD      | $2.1366 \pm 0.0161$ | $2.4095 \pm 0.0080$ | $2.6351 \pm 0.0179$ |
| RL-sBOED | $2.1257 \pm 0.0027$ | $2.3963 \pm 0.0333$ | $2.5307 \pm 0.0114$ |
| Step-DAD | $2.1471 \pm 0.0110$ | $2.4382 \pm 0.0048$ | $2.6621 \pm 0.0145$ |

TABLE III  
IEEE 14-BUS sPCE SCORES ( $L = 128$ ; NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.7541 \pm 0.0047$ | $0.8360 \pm 0.0297$ | $0.8464 \pm 0.0252$ |
| Fixed    | $4.2623 \pm 0.0402$ | $4.4946 \pm 0.0345$ | $4.6624 \pm 0.0083$ |
| Myopic   | $4.2907 \pm 0.0368$ | $4.3990 \pm 0.0177$ | $4.4641 \pm 0.0280$ |
| DAD      | $4.7576 \pm 0.0362$ | $4.8325 \pm 0.0047$ | $4.8443 \pm 0.0026$ |
| RL-sBOED | $4.7445 \pm 0.0022$ | $4.8275 \pm 0.0112$ | $4.8383 \pm 0.0108$ |
| Step-DAD | $4.7527 \pm 0.0275$ | $4.8297 \pm 0.0067$ | $4.8446 \pm 0.0042$ |

On IEEE 9-bus, DAD improves on Fixed by 0.1253, 0.0903, and 0.1286 nats at  $T = 3, 4, 5$ , respectively. Step-DAD adds 0.0105, 0.0287, and 0.0269 nats to the DAD mean. These are descriptive differences: separate method SDs do not quantify uncertainty in a paired comparison. Scores remain well below the 4.8598-nat estimator ceiling. The results favor the learned full-sequence policies over the tested fixed and myopic designs, but do not isolate the effect of feedback from policy parameterization and optimization.

On IEEE 14-bus, learned-policy means at  $L = 128$  lie close to the 4.8598-nat ceiling. Step-DAD changes the DAD mean by  $-0.0049$ ,  $-0.0028$ , and  $+0.0003$  nats at  $T = 3, 4, 5$ . These near-ceiling differences do not establish EIG rankings. Fixed exceeds Myopic at  $T = 4, 5$  in Table III, but this ordering reverses under larger- $L$  resampling (Section V-D).

#### B. Selected Probe Durations

Figure 2 summarizes duration distributions pooled across test episodes and, for fitted policies, training seeds. Fixed and the learned methods contribute 4608 histories each; Random and Myopic contribute 1536 each. Bars show interquartile ranges, not confidence intervals. At  $T = 5$ , final-stage medians are 0.369 s for DAD, 0.376 s for Step-DAD, 1.114 s for Fixed, 2.283 s for Myopic, and 2.979 s for Random. Sequential uniform sampling pushes Random's later durations toward the upper bound. RL-sBOED's final-stage medians are 2.585, 0.381, and 0.324 s for training seeds 101, 202, and 303, respectively; this variation contributes to its wide pooled ranges. All sequences obey the imposed ordering constraint, and different durations also imply different injected energies.

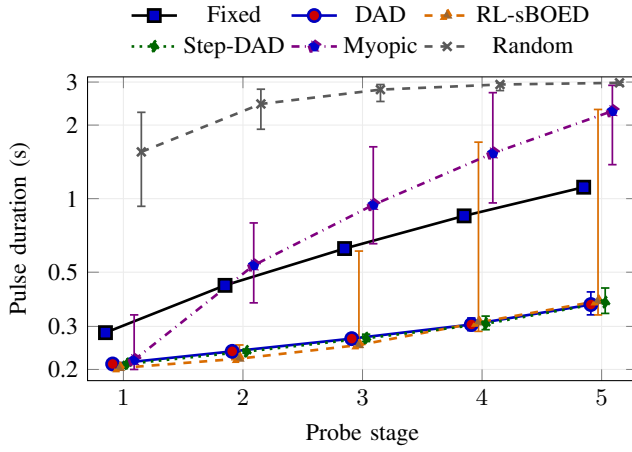


Fig. 2. IEEE 14-bus probe durations at  $T = 5$ . Markers show medians; bars span the 25th–75th percentiles. The vertical axis is logarithmic; horizontal offsets separate methods.

### C. Computational Cost

Table IV reports timings on NVIDIA A100 PCIe 40-GB GPUs, using training seed 101 and test seed 1001. Offline preparation includes initialization, optimization, and checkpoint validation; DAD and Step-DAD include both Fixed and DAD training. Online time averages the summed decision timers over 512 test episodes per method. It includes action selection and, where applicable, posterior assimilation and refinement, but excludes true-system propagation and information scoring. IEEE 9-bus profiling excludes one warm-up episode per method. Offline times are single-run measurements. All timed runs use  $L = 128$  for contrastive scoring, including training, validation, and Step-DAD refinement; Myopic uses 128 posterior particles and 32 simulated observations per candidate. Increasing  $L$  increases evaluation scoring time, which is excluded from the decision timers.

Let  $C_s$  denote one candidate-window simulation cost and  $C_\pi$  a policy forward pass. Dense  $m$ -machine RK4 integration over  $N$  time steps gives  $C_s = O(Nm^2)$ .

Online policy work is  $O(T)$  for Random/Fixed and  $O(TC_\pi)$  for DAD/RL-sBOED. Myopic uses  $O(TQC_e + TPC_s)$ , where  $P = 128$  particles,  $Q = 48$  candidate evaluations per decision, and  $C_e$  includes candidate simulation and likelihood calculations. Step-DAD refinement simulates  $UB(L+1)T(T-1)/2$  candidate windows before accounting for numerical Jacobians and conditional validation, with  $U = 4$ ,  $B = 32$ , and  $L = 128$ . Posterior assimilation adds  $O(TPC_s)$  work.

For  $J$  training updates, batch size  $B$ , and  $L$  contrasts, Fixed and DAD require  $O(JB(L+1)TC_g)$  simulation and gradient work, where  $C_g$  is the per-window cost. DAD’s reported preparation is  $W_F + W_D$ , and Step-DAD inherits both stages. RL-sBOED adds actor-critic replay updates to its simulated trajectory cost. Random and Myopic require no offline training. At IEEE 9-bus  $T = 5$ , Step-DAD’s mean score exceeds DAD’s by 0.0269 nats, while online decision time increases from 1.745 ms to 14.346 s per sequence. Thus,

TABLE IV  
COMPUTATIONAL COSTS FOR BOTH IEEE NETWORKS

| Method                             | IEEE 9-bus |         |         | IEEE 14-bus |         |         |
|------------------------------------|------------|---------|---------|-------------|---------|---------|
|                                    | $T = 3$    | $T = 4$ | $T = 5$ | $T = 3$     | $T = 4$ | $T = 5$ |
| Offline preparation (hours)        |            |         |         |             |         |         |
| Random                             | 0          | 0       | 0       | 0           | 0       | 0       |
| Fixed                              | 0.554      | 0.802   | 1.063   | 1.880       | 2.750   | 3.720   |
| Myopic                             | 0          | 0       | 0       | 0           | 0       | 0       |
| DAD                                | 1.116      | 1.608   | 2.139   | 3.810       | 5.457   | 7.418   |
| RL-sBOED                           | 0.100      | 0.117   | 0.129   | 0.163       | 0.205   | 0.249   |
| Step-DAD                           | 1.116      | 1.608   | 2.139   | 3.810       | 5.457   | 7.418   |
| Online decisions (ms per sequence) |            |         |         |             |         |         |
| Random                             | 0.167      | 0.229   | 0.303   | 0.158       | 0.207   | 0.285   |
| Fixed                              | 0.849      | 1.135   | 1.472   | 1.141       | 1.906   | 1.952   |
| Myopic                             | 4.418      | 5.952   | 7.514   | 9.998       | 13.433  | 16.918  |
| DAD                                | 1.012      | 1.345   | 1.745   | 1.440       | 1.513   | 1.733   |
| RL-sBOED                           | 1.147      | 1.781   | 7.692   | 1.413       | 1.550   | 1.905   |
| Step-DAD                           | 3.083      | 7.689   | 14.346  | 9.245       | 23.320  | 45.089  |

its modest information-score gain carries a substantial online cost in this implementation.

Rescoring  $n$  saved histories at a larger  $L$  requires  $O(n(L+1)TC_s)$  simulation work and  $O(nLT)$  likelihood arithmetic, without retraining. Measured decision times characterize algorithm execution rather than the real-time feasibility of the complete probing loop.

### D. Estimator Sensitivity

The ceiling  $\log(L+1)$  bounds the sampled sPCE score, not the true EIG. We rescore all 64 512 distinct IEEE 14-bus histories at  $L = 10^4$  and  $10^5$ , holding policies, generating parameters, durations, and observations fixed. No policies are retrained or rerun. Contrasts are drawn from the archived prior; matching test-seed/episode pairs share the contrast stream across methods, and the  $10^4$  samples are a prefix of the  $10^5$  samples. Archived terminal states and original scores are reproduced before rescoring. Tables V and VI report mean  $\pm$  SD using the seed aggregation defined in Section IV-B; seed-level summaries are included in the repository.

The estimator ceilings increase to 9.2104 and 11.5129 nats. The learned policies retain higher mean scores than the three baselines at both contrast counts. Myopic now exceeds Fixed at every horizon, showing that the longer-horizon ordering at  $L = 128$  was estimator-sensitive. At  $L = 10^5$ , Step-DAD changes the DAD mean by  $-0.0062$ ,  $-0.0082$ , and  $+0.0016$  nats at  $T = 3, 4, 5$ , respectively; refinement therefore provides no consistent mean-score improvement on this network. DAD’s  $T = 5$  mean increases from 8.666 to 9.693 nats between the two larger contrast counts. Thus, increasing  $L$  reduces the original ceiling effect but does not establish estimator convergence or a definitive EIG ranking among learned policies.

TABLE V  
IEEE 14-BUS SPCE SCORES ( $L = 10^4$ ; NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.8449 \pm 0.0121$ | $0.9552 \pm 0.0162$ | $0.9991 \pm 0.0388$ |
| Fixed    | $5.2956 \pm 0.0802$ | $6.0609 \pm 0.1246$ | $6.7906 \pm 0.0167$ |
| Myopic   | $5.8136 \pm 0.0514$ | $6.4715 \pm 0.0127$ | $6.8192 \pm 0.0500$ |
| DAD      | $7.1062 \pm 0.2327$ | $8.2061 \pm 0.0286$ | $8.6662 \pm 0.0556$ |
| RL-sBOED | $7.0793 \pm 0.0567$ | $8.1296 \pm 0.1355$ | $8.5852 \pm 0.2660$ |
| Step-DAD | $7.1008 \pm 0.2309$ | $8.1992 \pm 0.0191$ | $8.6583 \pm 0.0550$ |

TABLE VI  
IEEE 14-BUS SPCE SCORES ( $L = 10^5$ ; NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.8476 \pm 0.0136$ | $0.9645 \pm 0.0150$ | $1.0075 \pm 0.0333$ |
| Fixed    | $5.3472 \pm 0.0837$ | $6.1694 \pm 0.1401$ | $7.0488 \pm 0.0272$ |
| Myopic   | $5.9348 \pm 0.0697$ | $6.7628 \pm 0.0277$ | $7.2307 \pm 0.0540$ |
| DAD      | $7.3126 \pm 0.2674$ | $8.8217 \pm 0.0707$ | $9.6931 \pm 0.1308$ |
| RL-sBOED | $7.2873 \pm 0.0578$ | $8.7141 \pm 0.2070$ | $9.5528 \pm 0.4753$ |
| Step-DAD | $7.3064 \pm 0.2650$ | $8.8135 \pm 0.0578$ | $9.6947 \pm 0.1220$ |

### E. Scope and Limitations

The study uses reduced lossless dynamics, prescribed parameter priors, increasing durations, and one fixed actuator and endpoint sensor. Its information scores measure joint uncertainty reduction, not the accuracy of individual inertia or response estimates. Training and testing draw from the same prior; robustness to prior shift, sensor bias, and unmodeled dynamics is not assessed. Finite-contrast scoring, posterior particles, and ODE integration introduce distinct approximations. The simulator also assumes zero decision latency, so these results do not establish real-time deployment or improved control safety.

## VI. CONCLUSION

We formulated sequential duration selection for power-system probing with uncertain inertia and effective frequency response, retaining the physical state between experiments. On the reduced IEEE 9-bus and IEEE 14-bus benchmarks, learned adaptive policies achieve higher mean sPCE information scores than the tested random, fixed, and myopic designs at equal probe counts. DAD and RL-sBOED shift optimization offline and use policy evaluation online. Step-

DAD offers modest additional gains on IEEE 9-bus at substantially greater online cost; IEEE 14-bus rescoring preserves the learned-policy advantage while exposing estimator-sensitive baseline rankings and no consistent refinement benefit. Residual contrast-count sensitivity limits fine EIG rankings. The findings support learned sequential design within this benchmark and motivate further study of model mismatch, sensing constraints, and estimator accuracy.

## REFERENCES

- [1] P. Denholm, T. Mai, R. W. Kenyon, B. Kroposki, and M. O'Malley, "Inertia and the power grid: A guide without the spin," National Renewable Energy Laboratory, Tech. Rep. NREL/TP-6A20-73856, 2020. [Online]. Available: <https://docs.nrel.gov/docs/fy20osti/73856.pdf>
- [2] P. Wall and V. Terzija, "Simultaneous estimation of the time of disturbance and inertia in power systems," *IEEE Transactions on Power Delivery*, vol. 29, no. 4, pp. 2018–2031, 2014.
- [3] J. W. Pierre, N. Zhou, F. K. Tuffner, J. F. Hauer, D. J. Trudnowski, and W. A. Mittelstadt, "Probing signal design for power system identification," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 835–843, 2010.
- [4] J. Peng, J. Tan, P. Koralewicz, E. Mendiola, S. Dong, A. Hoke, and K. Horowitz, "Probing signal-based inertia and frequency response estimation for power systems with high levels of inverter-based resources," in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [5] H. Hjalmarsson, "System identification of complex and structured systems," *European Journal of Control*, vol. 15, no. 3–4, pp. 275–310, 2009.
- [6] X. Huan and Y. M. Marzouk, "Sequential Bayesian optimal experimental design via approximate dynamic programming," arXiv:1604.08320, 2016. [Online]. Available: <https://arxiv.org/abs/1604.08320>
- [7] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, "Deep adaptive design: Amortizing sequential Bayesian experimental design," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395.
- [8] T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli, "Optimizing sequential experimental design with deep reinforcement learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 2107–2128.
- [9] M. Hedman, D. R. Ivanova, C. Guan, and T. Rainforth, "Step-DAD: Semi-amortized policy-based Bayesian experimental design," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 22 904–22 923. [Online]. Available: <https://proceedings.mlr.press/v267/hedman25a.html>
- [10] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MAT-POWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [11] R. Kaur and D. Kumar, "Transient stability improvement of IEEE 9 bus system using power world simulator," *MATEC Web of Conferences*, vol. 57, p. 01026, 2016.